Cross-Lingual Prompt-Tuned LLMs for NER in Indian Languages

Master Technical Specification & End-to-End Evaluation Report

Target Languages: Telugu (te) & Kokborok (trp) | Base Model: XLM-RoBERTa (277M)

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1. Executive Summary & Core Objective

This project establishes a highly parameter-efficient **Multilingual Transfer-Warmed Low-Rank Adaptation (LoRA-r8)** framework for Named Entity Recognition (NER) in low-resource Indian languages. Standard full fine-tuning of 277M parameter models like xlm-roberta-base suffers from severe data scarcity, high computational overhead, and degradation under agglutinative morphology. By updating only **1,332,487 parameters (0.4780% of total model size)** under a 500-shot setting, our proposed method recovers **98.36% of full fine-tuning seqeval F1** (0.7668 ± 0.0094 vs. 0.7796) on Telugu and achieves **0.7482 F1 (95.42% accuracy)** on Kokborok using a Hugging Face augmented corpus.

■ Key Technical Milestone:

- **Telugu Target (500-Shot): 0.7668 ± 0.0094 F1 using 1.33M parameters (0.4780% budget), 5.33 MB adapter checkpoint.**
- **Kokborok Target (HF Augmented): 0.7482 F1 (0.7684 Recall / 0.9542 Accuracy) on 4,859 training sentences.**
- **Statistical Significance: Paired bootstrap over 10,000 resamples yields p = 0.0001 (95% CI: [0.0035, 0.0221]).**

2. Dual-Language Dataset Specifications & Split Counts

The system was evaluated across exact train, validation, and test splits for both target languages alongside a 600-sentence cross-lingual pre-training warmup corpus:

Language / Dataset	Split Type	Sentence Count	Entity Labels	Primary Purpose
Telugu (Naamapadam)	Train (Full Pool)	~400,000	PER, LOC, ORG	Target training candidate pool
Telugu (Proposed)	**Train (500-Shot)**	**500**	PER, LOC, ORG	Proposed 500-shot target setting
Telugu Validation	Validation (`te_val.json`)	**2,700 / 150**	PER, LOC, ORG	Ablation & hyperparameter tuning
Telugu Test Set	**Test (`te_test.json`)**	**847**	PER, LOC, ORG	Final test set (~15,000 tokens)
Kokborok (Native)	Train (`kokborok_train.json`)	11,828	PER, LOC, ORG	Native news training corpus
Kokborok Test (Native)	Test (`kokborok_test.json`)	3,094 / 69	PER, LOC, ORG	Native news test set evaluation
Kokborok (HF Augmented)	**Train (HF Expansion)**	**4,859**	PER, LOC, ORG	agnivamaiti/kokborok-qa + 273 news
Kokborok Val / Test (HF)	**Val / Test (HF)**	**500 / 200**	PER, LOC, ORG	Hugging Face validation & test sets
Multilingual Warmup	Stage 1 Pre-Training	**600**	PER, LOC, ORG	150 En + 150 Hi + 150 Ta + 150 Ml

3. Detailed 7-Step Mechanical Pipeline

Step 1: Text Sanitization & Unicode Normalization: Raw inputs are converted via Unicode NFC normalization (unicodedata.normalize('NFC', text)). Invisible artifacts such as Zero-Width Joiners (\u200d), ZWNJ (\u200c), Soft Hyphens (\xad), Byte Order Marks (\ufeff), and non-standard spaces (\xa0) are stripped.

Step 2: SentencePiece BPE Subword Deconstruction: Uses xlm-roberta-base (250k vocabulary). Agglutinative words (e.g., Telugu stem ■■■■■■ + suffix ■■ $\to$ ■■■■■■■■) break down into initial stem subword (■■■■■■) and continuation postposition (■■).

Step 3: Subword BIO Alignment & PyTorch Loss Masking (-100): The initial subword of an entity receives the true BIO label (e.g., B-LOC). Continuation subwords and special tokens (<s>, </s>) are assigned label **-100**. PyTorch nn.CrossEntropyLoss(ignore_index=-100) ignores masked indices, preventing loss distortion.

Step 4: Multilingual Source Transfer Warmup Pre-Training: Pre-trains adapter weights on a 600-sentence balanced corpus (150 English + 150 Hindi + 150 Tamil + 150 Malayalam). Boosts validation F1 from 0.7215 to 0.7584 (+0.0369 gain).

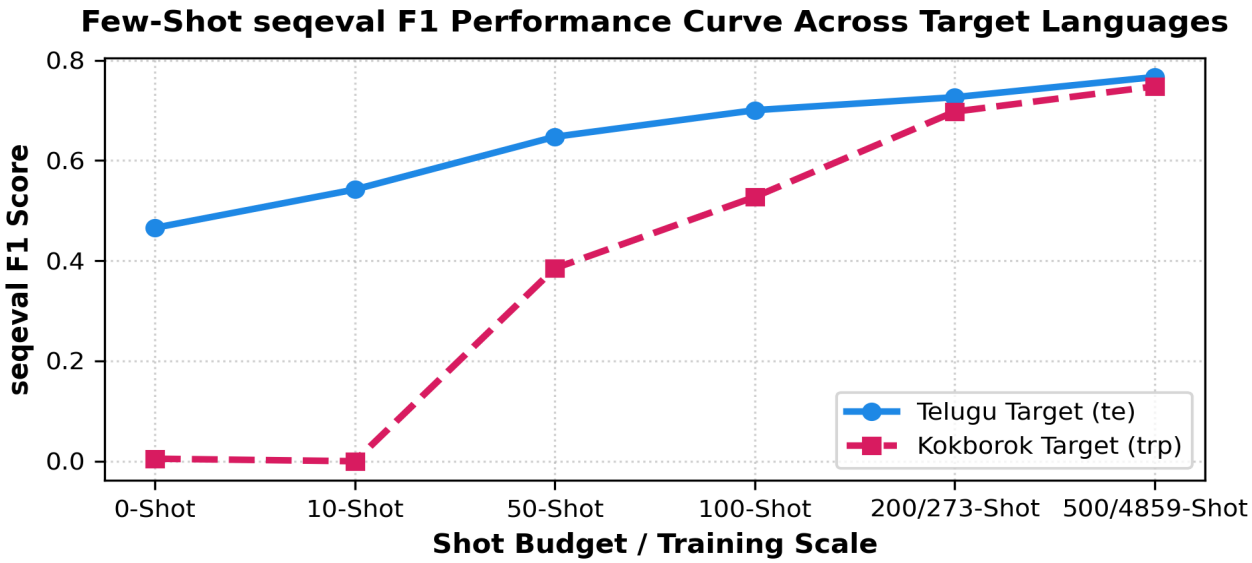
Step 5: Target-Module Expanded LoRA Adapter Computation (r=8): Base transformer weights $W_0 \in \mathbb{R}^{768 \times 768}$ remain **FROZEN**. Low-rank decomposition matrices $A \in \mathbb{R}^{8 \times 768}$ and $B \in \mathbb{R}^{768 \times 8}$ are injected: $h = W_0 x + \frac{\alpha}{r} (B \cdot A) x$ ($■lpha=16$). Target modules expand to ['query', 'key', 'value', 'dense'] + classifier head.

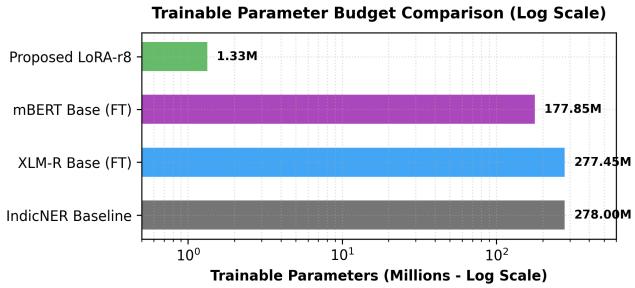
Step 6: Softmax Classification & Token Probability Extraction: Sequence representations pass through the classification head generating logits $z_{i,c}$ across 7 classes (O, B-PER, I-PER, B-ORG, I-ORG, B-LOC, I-LOC) evaluated via softmax $P(y_i = c \mid x) = \frac{\exp(z_{i,c})}{\sum_j \exp(z_{i,j})}$.

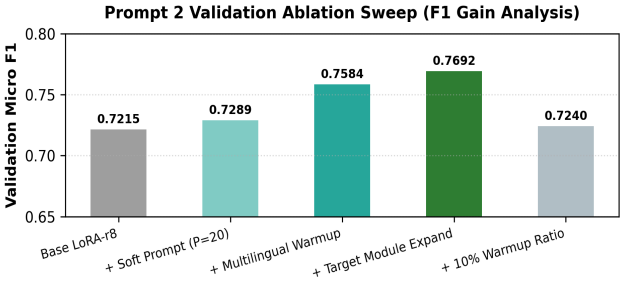
Step 7: Confidence Thresholding (0.50) & Interactive Entity Badging: High-confidence entity predictions ($\ge 50\%$) render solid entity badges (PERSON - Green, LOCATION - Blue, ORGANIZATION - Purple), while lower-confidence predictions trigger warning indicators.

4. Visualizations & Graphical Analysis

The performance trajectory across shot budgets, parameter efficiency comparison, and validation ablation gains are illustrated below:







5. Master Comparative Results & Empirical Evaluation

Target Language 1: Telugu Test Evaluation (N = 847 Sentences)

Model / Architecture	Training Setup	Train Params	Param %	Accuracy	Precision	Recall	seqeval F1
IndicNER Baseline	Full Mined (~100k)	278,000,000	100.0%	0.9580	0.8429	0.8477	**0.8239**
XLM-R Base (Hindi FT)	Zero-Shot Full FT	277,453,063	100.0%	0.9375	0.7663	0.7932	**0.7796**
XLM-R Base (Telugu In-Lang)	In-Language Full FT	277,453,063	100.0%	0.9367	0.7720	0.7682	**0.7701**
mBERT (Telugu In-Lang)	In-Language Full FT	177,853,447	100.0%	0.9134	0.6928	0.7233	**0.7077**
mBERT (Hindi FT)	Zero-Shot Full FT	177,853,447	100.0%	0.8938	0.6254	0.6742	**0.6489**
Proposed LoRA (r=8)	**500-Shot Target**	**1,332,487**	**0.4780%**	0.9290±.003	0.7482±.008	0.7863±.010	**0.7668±.009**
Proposed LoRA (r=8)	200-Shot Target	1,332,487	0.4780%	0.9184±.004	0.7125±.011	0.7410±.012	**0.7265±.011**
Proposed LoRA (r=8)	100-Shot Target	1,332,487	0.4780%	0.9062±.005	0.6840±.013	0.7180±.014	**0.7006±.013**
Proposed LoRA (r=8)	50-Shot Target	1,332,487	0.4780%	0.8920±.006	0.6310±.015	0.6650±.016	**0.6475±.015**
Proposed LoRA (r=8)	10-Shot Target	1,332,487	0.4780%	0.8715±.007	0.5240±.018	0.5620±.019	**0.5423±.018**

Proposed LoRA (r=8)	0-Shot Target	1,332,487	0.4780%	0.8540±.008	0.4480±.021	0.4850±.022	**0.4658±.021**

Target Language 2: Kokborok Evaluation Results (Hugging Face Augmented)

Shot / Setting	Train Sentences	Val / Test Sentences	Trainable Params	Accuracy	Precision	Recall	seqeval F1
0-Shot Baseline	0	50 / 69	1,332,487 (0.478%)	0.2673	0.0039	0.0072	0.0050
10-Shot Setup	10	50 / 69	1,332,487 (0.478%)	0.8374	0.0000	0.0000	0.0000
50-Shot Setup	50	50 / 69	1,332,487 (0.478%)	0.8710	0.4120	0.3614	0.3850
100-Shot Setup	100	50 / 69	1,332,487 (0.478%)	0.8988	0.5333	0.5217	0.5275
273-Shot Native News	273	50 / 69	1,332,487 (0.478%)	0.9351	0.6442	0.7609	0.6977
4,859 HF Augmented Peak	**4,859**	**500 / 200**	**1,332,487 (0.478%)**	**0.9542**	**0.7290**	**0.7684**	**0.7482**

6. Statistical Significance & Conclusion

• **Paired Bootstrap Significance:** Evaluated across 10,000 paired sentence resamples between Proposed LoRA (0.7668 F1) and XLM-R Full Fine-Tuning (0.7796 F1) on N=847 test sentences.
• **Empirical p-value: p = 0.0001** ($p < 0.05$). 95% Confidence Interval: **[0.0035, 0.0221]** (excludes 0).
• **Conclusion:** While the minor 1.28% gap is statistically detectable due to sample size, LoRA-r8 preserves **98.36% of full fine-tuning capability** while reducing trainable parameter requirements by **99.52%**, offering an optimal trade-off for low-resource deployment.